

SRI CHAITANYA IIT ACADEMY, INDIA

AP • TELANGANA • KARNATAKA • TAMILNADU • MAHARASHTRA • DELHI • RANCHI

PRAGYA G8 – ACHIEVERS | MATHEMATICS
ZERO REPORT | INTEGRATED PAPER – II (JEE ADVANCED)

Paper	Integrated Paper – II, JEE Advanced pattern (file: Paper_II_ADVANCED.pdf)
Batch	PRAGYA G8 – Achievers
Syllabus	Surds; Logarithms; Sequences & Series; Quadratic Equations
Pattern	18 questions – Section I: Q1–8 integer (0–9); Section II: Q9–14 one or more correct; Section III: Q15–18 single correct
Date of report	02-10-2026
Result	All 18 keys verified. 5 corrections made (listed below). Outstanding errors: 0.

ANSWER KEY

1	2	3	4	5	6
4	5	2	1	9	7
7	8	9	10	11	12
6	3	(A, B, C)	(A, B, D)	(B, C, D)	(A, C, D)
13	14	15	16	17	18
(A, B)	(A, C)	(D)	(A)	(B)	(C)

SUMMARY

Section I answers	4, 5, 2, 1, 9, 7, 6, 3 – all single digits, no repeats
Section II keys	two items with 2 correct, four with 3 correct; (A) correct in 5, (B) 4, (C) 4, (D) 3
Section III keys	(D), (A), (B), (C) – one of each
Difficulty	Easy 9 Medium 7 Difficult 2
Chapter usage	Surds 11 Logarithms 10 Seq. & Series 16 Quadratic Eqns 13 (questions touching each chapter)
Syllabus / method level	Within G8 CPT-01 scope; no excluded topic (AGP, sigma, V_n , telescopic series, location of roots, log inequalities, graphs); no calculus, Cauchy–Schwarz, Newton sums, cubic equations, trigonometry or GIF.

QUESTION-WISE ANALYSIS

Q	Type	Chapters	Concept tested	Level	Key
1	INT	Surds, QE, S&S	Cube root of $a + b\sqrt{2}$ as a binomial surd	M	4
2	INT	S&S	AP–GP–HP chain $\Rightarrow c^2 = ae$	D	5
3	INT	Surds, QE, S&S, Logs	Infinite nested radical $\Rightarrow$ quadratic	E	2
4	INT	Logs, Surds	Characteristic; digits and leading zeros	M	1
5	INT	QE, Surds, S&S	$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$	M	9
6	INT	Logs, QE, S&S	Quadratic in $\log_5 x$	E	7
7	INT	S&S, Surds, QE	GP terms; product of roots	E	6
8	INT	QE, S&S, Logs	$f^8 = 1$: base 1 / index 0 / base -1 with even index	M	3
9	MCQ+	Surds, QE, S&S	Denesting; sum/product; AP test	E	(A, B, C)
10	MCQ+	S&S	Infinite GP of squares and cubes	M	(A, B, D)
11	MCQ+	Logs, QE, S&S	Quadratic in $\log_3 x$	E	(B, C, D)
12	MCQ+	Surds, QE, S&S	Denesting; sum/product; GP test	E	(A, C, D)
13	MCQ+	QE, Surds, S&S, Logs	Roots $3 \pm \sqrt{2}$; AP and log tests	E	(A, B)
14	MCQ+	Logs, S&S	$\log_N a, \log_N b, \log_N c$ in AP $\Rightarrow$ reciprocals in HP	M	(A, C)
15	SCQ	Logs, QE, Surds	Identical equations $\Rightarrow \log_a b = \log_b a = \pm 1$	D	(D)
16	SCQ	S&S, Surds, Logs	HM of conjugate surds; $7 + 4\sqrt{3} = (2 + \sqrt{3})^2$	M	(A)
17	SCQ	QE, Surds, S&S	HM = $2\alpha\beta/(\alpha + \beta)$; rationalising	E	(B)
18	SCQ	Logs, QE, S&S	Quadratic in $\log_2 x$; GM	E	(C)

INT = integer answer; MCQ+ = one or more correct; SCQ = single correct; E/M/D = easy/medium/difficult.

ERRORS FOUND AND CORRECTED

- Title / header.** “Hell Paper” and “(HELL LEVEL)” removed. Paper is now *Integrated Paper – II, JEE Advanced pattern*; running header reads *Paper II – Advanced*.
- Tagline removed.** “Every question draws on at least three of ...” was not true: Q2 and Q10 use Sequences & Series only; Q4 uses Logarithms and Surds; Q14 uses Logarithms and Sequences & Series.
- Instruction and marking-scheme boxes removed** from all three sections. The Section I box was also wrong for an integer section (“+3 if ONLY the correct *option* is chosen” – the section has no options).
- Q14 – objection route closed.** The statement allowed a constant progression ($a = b = c$), for which (B) and (D) are also true. Now reads “Let a, b, c be *distinct* positive reals ...”. Key unchanged: (A), (C).
- Page break.** Q12 had options (C) and (D) printed on the next page. Every question is now kept on one page with its options; the two equations in Q15 are set on their own line.

No answer in the paper was wrong. Every Section II item has at least one correct option and no two identical options; every Section III item has exactly one correct option.

OBSERVATIONS (NO CHANGE MADE)

- Q8 is the same construction as Paper I Q2 ($f(x)^{g(x)} = 1$ with the base = -1 case) on different numbers; a student who has written Paper I will recognise it. Omitting the base = -1 case gives $T = 20$ and no integer answer, so there is no rival key.
- Q1 is meant to be done by spotting $7 \pm 5\sqrt{2} = (1 \pm \sqrt{2})^3$ (cube of a binomial surd, as in the Surds L1/L2 cube-root items). The other common route – cubing $s = u + v$ – leads to $s^3 + 3s - 14 = 0$, solvable only by inspecting $s = 2$; borderline for the no-cubics rule.
- Redundant but harmless conditions: “positive” in Q5 (the ratio is forced to be 9) and “whose other root is also a term” in Q7 (the other root is $2 = a_2$ automatically).
- Repeated templates: Q9 and Q12 (denest $\sqrt{m \pm 2\sqrt{n}}$, then sum/product/progression/quadratic); Q6, Q11 and Q18 (quadratic in a logarithm, then a GP).
- Q10 (C) is a distractor: the sum of the cubes is $\frac{375}{7}$, not $\frac{375}{13}$.

KEY VERIFICATION

- Q1 4** $7 \pm 5\sqrt{2} = (1 \pm \sqrt{2})^3 \Rightarrow u + v = 2, uv = -1$; roots $2, -1 \Rightarrow p = -1, q = -2$; $r = q^2/p = -4, |r| = 4$
- Q2 5** $b = \frac{a+c}{2}, d = \frac{c^2}{b} = \frac{2ce}{c+e} \Rightarrow c(c+e) = (a+c)e \Rightarrow c^2 = ae = 25, c = 5$
- Q3 2** $x^2 = x + 12 \Rightarrow x = 4; y^2 = y + 72 \Rightarrow y = 9; g = 6; \log_6 36 = 2$
- Q4 1** $\log 6^{10} = 7.781 \Rightarrow d = 8; \log 6^{-10} = \bar{8}.219 \Rightarrow z = 7; d - z = 1$
- Q5 9** $(\alpha - \beta)^2 = 9 + 4\sqrt{5} - 4\sqrt{5} = 9 \Rightarrow \alpha - \beta = 3; \frac{1}{3}r = 3 \Rightarrow r = 9$
- Q6 7** $t^2 - 3t + 2 = 0 \Rightarrow x = 5, 25; y = 25^2/5 = 125; 3 + 4 = 7$
- Q7 6** $r^6 = 8 \Rightarrow r = \sqrt{2}, a_4 = 4$; other root $8/4 = 2 = a_2; k = 6$
- Q8 3** base 1: $x = 2, 5$; index 0: $x = 6, 7$; base -1 : $x = 3, 4$ (index 12, 6 even); $T = 27; 3r^2 = 27 \Rightarrow r = 3; \log_3 27 = 3$
- Q9 (A, B, C)** $a = \sqrt{5} + \sqrt{3}, b = \sqrt{5} - \sqrt{3}; a + b = 2\sqrt{5}, ab = 2$; AP needs $a + b = 4$ – false
- Q10 (A, B, D)** $\frac{1+r}{1-r} = \frac{25}{15} \Rightarrow r = \frac{1}{4}, a = \frac{15}{4}$; cubes: $\frac{a^3}{1-r^3} = \frac{375}{7} \neq \frac{375}{13}; ar^2 = \frac{15}{64}$
- Q11 (B, C, D)** $t = 1, 3 \Rightarrow x = 3, 27; \log \text{sum} = 4$ (A false); $x_1 x_2 = 81 = 9^2; 27 - 3 = 24$
- Q12 (A, C, D)** $p = 3 + \sqrt{5}, q = 3 - \sqrt{5}; p + q = 6, pq = 4$ (B false); $2^2 = pq; x^2 - 6x + 4 = 0$
- Q13 (A, B)** $\alpha, \beta = 3 \pm \sqrt{2}; \alpha - \beta = 2\sqrt{2}, \alpha\beta = 7; 14 \neq 6 + 2\sqrt{2}; \log_\alpha \beta = -1 \iff \alpha\beta = 1$ – false
- Q14 (A, C)** $b^2 = ac \Rightarrow \log_N a, \log_N b, \log_N c$ in AP, so $\log_a N, \dots$ in HP; $(b^2)^2 = a^2 c^2$; distinct terms cannot be in AP and HP together, nor a GP in AP
- Q15 (D)** $t = 1/t \Rightarrow t = \pm 1; t = 1: x^2 + x + 1$ (not real); $t = -1: b = 1/a, x^2 - x - 1 = 0, \alpha - \beta = \sqrt{5}, ab = 1$
- Q16 (A)** $b = \frac{2ac}{a+c} = \frac{2}{14} = \frac{1}{7}; \log_{2+\sqrt{3}}(2 + \sqrt{3})^2 = 2; k = \frac{1}{2}(\frac{1}{7} + 2) = \frac{15}{14}$

Q17 (B) roots $\sqrt{5}, \sqrt{3}; \frac{2\sqrt{15}}{\sqrt{5}+\sqrt{3}} = \sqrt{15}(\sqrt{5} - \sqrt{3}) = 5\sqrt{3} - 3\sqrt{5}$

Q18 (C) $t^2 - 2t - 8 = 0 \Rightarrow t = 4, -2 \Rightarrow x = 16, \frac{1}{4}; g = \sqrt{4} = 2$

VERIFICATION METHOD

1. **Brute force from the statement** – equations solved by numerical scanning, integer conditions by exhaustive search; every Section II option evaluated as a statement (Q14 tested on 2000 random progressions).
2. **Exact arithmetic along the hand method** – fractions and exact $a + b\sqrt{n}$ numbers; 60-digit decimals where a logarithm is unavoidable.
3. **Build-time check** – keyed options re-matched to the true values before typesetting.
4. **Independent re-solve** – a separate solver, given only the printed paper, solved all 18 from scratch: all 18 answers matched the key (Q14 also tested on 100,000 random progressions); no wrong option, duplicate option, typo, objectionable reading or layout defect.